Photoelectric effect and determination of Planck's constant

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1 Introduction

When a metal is subject to electromagnetic radiation, the electrons in the metal may be excited by incident photons to the point that they leave the metal in a phenomenon known as the *photoelectric effect*, which was first explained theoretically in the first of Einstein's 1905 *annus mirabilis* papers. This explanation is bound up in the history of another phenomenon, *blackbody radiation*, of which the investigation by Max Planck first introduced the notion that energy may be quantized (though Planck was hesitant to assign physical significance to their findings).

1.1 Blackbody radiation

Everything with a temperature above absolute zero, as it turns out, emits thermal electromagnetic radiation. This is most readily observed in things like red-hot metal, thermal imaging cameras, and sunlight.

1.1.1 Rayleigh and Jeans

Einstein built off the results of Planck, who in turn had corrected the results of Jeans and Rayleigh.

Rayleigh and Jeans sought to model the behavior of a *blackbody*, a hypothetical object that absorbs all incident light, which they could approximate with a *Jeans cube*, a hollow, metal cube with a small hole, such that the hole functions as a blackbody.

One dimension Such a cube, if heated, would give rise to standing electromagnetic waves along the cube's 3 axes, as the light emitted from the walls would reflect back onto the incoming wave with a 180° phase shift. Such a standing wave, in one dimension, is described by

$$E(x,t) = 2E_0 \sin(kx)\cos(\omega t)$$

for a wavenumber $k = 2\pi/\lambda$.

But these waves are subject to the boundary condition that $E = 0$ at the walls, as any electric field there would immediately be cancelled by the movement of the charge of the wall itself. For a cube of side length L, this implies

$$\begin{aligned}
&sin(kL) = 0 \\
&\implies kL = n\pi \\
&\implies \frac{n\pi}{L} = \frac{2\pi}{\lambda} \\
&\implies \lambda = \frac{2L}{n} \qquad \forall n \in \mathbb{N},
\end{aligned}$$

or in terms of frequency,

$$\nu = \frac{nc}{2L}.$$

This result limits the frequencies of light possible, at which point it makes sense to talk about the *density of states*

$$\rho = 2\frac{n}{\nu} = \frac{4L}{c},$$

such that $\rho\Delta\nu$ is the number of states in a range of frequencies, and where the factor of two accounts for the polarization of the wave.

This result is only one dimensional. It must now be generalized to three dimensions.

Three dimensions Radiation of some frequency ν inside the cube would be decomposable into three components along each axis of the cube, subject to previous results, that is,

$$\begin{aligned}
&\mathbf{E}(\mathbf{r}, t) = \begin{pmatrix} E_x(x,t) \\ E_y(y,t) \\ E_z(z,t) \end{pmatrix} = \begin{pmatrix} 2E_{0x}\sin(2\pi x/\lambda_x)\cos(\omega t) \\ 2E_{0y}\sin(2\pi y/\lambda_y)\cos(\omega t) \\ 2E_{0z}\sin(2\pi z/\lambda_z)\cos(\omega t) \end{pmatrix} \\
&\implies \begin{cases} \lambda_x = 2L/n_x \\ \lambda_y = 2L/n_y \\ \lambda_z = 2L/n_z \end{cases} \\
&\implies \lambda = \sqrt{\lambda_x^2 + \lambda_y^2 + \lambda_z^2} \\
&\implies \nu = \frac{c}{\lambda} = \frac{c}{2L}\sqrt{n_x^2 + n_y^2 + n_z^2}.
\end{aligned}$$

Rayleigh and Jeans proceeded by calculating the number of states per range of frequencies, and then applying a result from statistical mechanics in order to arrive at the average total energy.

In the one dimensional case, this number of states was calculable by multiplying the density of states, which was constant and easily obtained, by the length of the range of frequencies. In the three-dimensional case, it is not immediately apparent what this density is.

A geometric approach yields the answer. By treating the numbers n_x, n_y, and n_z as describing points in $\mathbb{N}^3$, it makes sense to treat the quantity $\sqrt{n_x^2 + n_y^2 + n_z^2}$ as a radius. This means that, for some frequency ν, the set of points in $\mathbb{N}^3$ that lie on a sphere of radius $n = \frac{2L}{c}\nu$ describes all the possible combinations of n_x, n_y, and n_z for that frequency.

This is not in itself a useful result, as the intersection of such a sphere and $\mathbb{N}^3$ is empty for most frequencies. Instead, a range of frequencies $\mathrm{d}\nu$ must be considered, and therefore the intersection of a shell of thickness $\mathrm{d}r = \frac{2L}{c}\mathrm{d}\nu$ with $\mathbb{N}^3$. The size of this intersection must be equal to the volume of the part of the shell that lies in $\mathbb{R}^3_+$ times the volumetric density of points in $\mathbb{N}^3$, which may be seen trivially to equal 1:

$$\begin{aligned}
\mathrm{d}N(r) &= \rho_{\mathbb{N}}\mathrm{d}V(r) = \frac{1}{8}4\pi r^2\mathrm{d}r \\
= \mathrm{d}N(\nu) &= \frac{1}{2}\pi\left(\frac{2L}{c}\right)^3\nu^2\mathrm{d}\nu \\
\implies 2\frac{\mathrm{d}N}{\mathrm{d}\nu} &= \frac{8\pi L^3}{c^3}\nu^2 = \rho(\nu).
\end{aligned}$$

Energy density Rayleigh and Jeans wanted ultimately to derive an expression for spectral energy density (energy per unit volume per unit wavelength), as that was the quantity that could be measured empirically for an actual Jeans cube with which they wanted to compare their model.

To this end, the pair treated the system as a classical thermodynamic system, such that the probability of each state follows the Boltzmann distribution and the average energy per state may be expressed

$$\begin{aligned}
\bar{E} &= \sum \text{energy} \times \text{probability} \\
&= \sum_i \varepsilon_i p_i \\
&= \sum_i \varepsilon_i \left(Z^{-1}\exp\left(-\frac{\varepsilon_i}{k_\mathrm{B}T}\right)\right) \\
&= \frac{\sum_i \varepsilon_i \exp\left(-\frac{\varepsilon_i}{k_\mathrm{B}T}\right)}{\sum_j \exp\left(-\frac{\varepsilon_j}{k_\mathrm{B}T}\right)},
\end{aligned} \tag{1}$$

at which point the assumption is made in classical physics that energy levels are continuous, that is, energy may take on any real value. Here, a justification can be made (as is often) to turn the summations into integrals with respect to E:

$$\begin{aligned}
\bar{E} &= \frac{\int_0^\infty E\exp\left(-\frac{E}{k_\mathrm{B}T}\right)\mathrm{d}E}{\int_0^\infty \exp\left(-\frac{E}{k_\mathrm{B}T}\right)\mathrm{d}E} \\
&= \frac{(k_\mathrm{B}T)^2}{k_\mathrm{B}T} = k_\mathrm{B}T,
\end{aligned}$$

which is just the result of the equipartition theorem from statistical mechanics.

Spectral energy density may now be given as

$$\begin{aligned} u_T(\nu) &= \frac{\text{energy}}{\text{state}} \times \frac{\text{states}}{\text{unit frequency}} / \text{volume} \\ &= \bar{E}\rho(\nu)/L^3 \\ &= \frac{8\pi k_{\mathrm{B}} T}{c^3}\nu^2, \end{aligned} \tag{2}$$

which is the *Rayleigh-Jeans law*.

Though this result agreed well with data at low frequencies, it predicted spectral energy density to go to infinity with increasing frequency, which was completely unphysical and at odds with empirical data and the theoretical results of Boltzmann as given by the Stefan-Boltzmann law.

1.1.2 Wien

The work of Wilhelm Wien was also instrumental in the formulation of Planck's law.

1.1.3 Planck

At this point, there existed two laws that were valid at opposite ends of the electromagnetic spectrum. Max Planck recognized this and "patched" the two together, arriving an equation of the form

$$u_T(\nu) = \frac{A}{\exp(B\nu/T) - 1}\nu^3, \tag{3}$$

which could be fit to empirical data remarkably well across the entire spectrum.

At this point, Planck would make the theoretical leap that would lead to the development of quantum physics by assuming energy was discrete rather than continuous, and uniformly distributed over every state, that is, $\varepsilon_n = n\varepsilon$, for some fundamental unit of energy ε. With these changes, Equation 1 then becomes

$$\bar{E} = \frac{\varepsilon \sum_i i \exp\left(-\frac{i\varepsilon}{k_{\mathrm{B}}T}\right)}{\sum_j \exp\left(-\frac{j\varepsilon}{k_{\mathrm{B}}T}\right)},$$

whose sums converge, giving

$$\bar{E} = \frac{\varepsilon}{\exp\left(-\frac{\varepsilon}{k_{\mathrm{B}}T}\right) - 1}$$

and therefore

$$u_T(\nu) = \frac{8\pi}{c^3}\frac{\varepsilon}{\exp\left(-\frac{\varepsilon}{k_{\mathrm{B}}T}\right) - 1}\nu^2.$$

In order to reconcile this with the results of Rayleigh, Jeans, and Wien, and Equation 3, however, ε had to increase with frequency, to recover $\bar{E} \approx k_{\mathrm{B}}T$ at low frequencies and $\bar{E} \approx 0$ at high frequencies. These conditions implied

$$\varepsilon = h\nu, \tag{4}$$

with a proportionality constant of h and thence

$$u_T(\nu) = \frac{8\pi h}{c^3} \frac{1}{\exp\left(-\frac{h\nu}{k_{\mathrm{B}}T}\right) - 1} \nu^3, \tag{5}$$

known as *Planck's law*.

1.2 Photoelectric effect

Planck struggled to make sense of these results, and attributed this quantization to some property of the matter in the cube. It took Albert Einstein, in attempting to explain the photoelectric effect, to posit that quantization was attributable to the light itself, which was not exactly made up of waves as Maxwell had it, but quanta called photons, each with energy given by Equation 4 (which would become known as the *Planck-Einstein relation*).

Einstein modeled electrons on the surface of the metal as residing in a potential energy well due to the lattice structure of the material, with the depth of the well ϕ known as the *workfunction* of the material. When light with energy $h\nu > \phi$ is shone upon the metal, the light's photons may imbue the metal's electrons with enough energy to escape with a kinetic energy of up to $h\nu - \phi$.

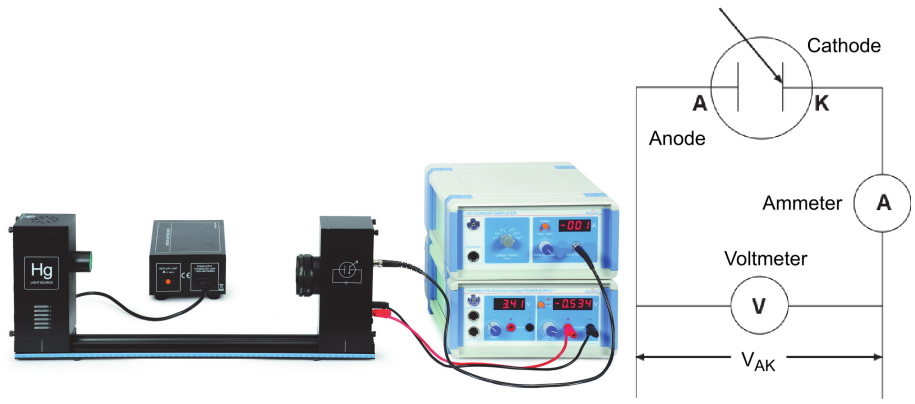
If the metal the light is incident upon is the cathode of a photodiode, then any electrons emitted may be picked up by the anode and read as a current. If a potential difference is applied across the photodiode, the voltage at which no current is read for light of a specific frequency becomes the *stopping voltage*

$$V_{\mathrm{stop}} = K/e = \frac{h}{e}\nu - \frac{\phi}{e}. \tag{6}$$

2 Materials and Procedure

The experiment makes use of the PASCO Scientific Photoelectric Effect apparatus (Figure 1), which includes a mercury vapor lamp and photodiode circuitry enclosure mounted on a track. The electrical system consists of a power supply for the lamp, DC current amplifier that serves as a high sensitivity ammeter, and a DC voltage power supply for applying stopping voltages. Optical parameters are controlled using a set of five wavelength filters (365 nm, 405 nm, 436 nm, 546 nm and 577 nm) and varying aperture rings (2 mm, 4 mm and 8 mm) to adjust light intensity.

The mercury lamp was allowed to warm up for ten minutes after activation. During this time, the current amplifier was zeroed by placing a cover over the photodiode and adjusting the zeroing knob until the meter read 0 A.



(a) Experimental setup. (b) Circuit diagram.

Figure 1: PASCO Scientific Photoelectric Effect apparatus.

The stopping voltage was measured for each of the five wavelengths. For each filter, the DC power supply was adjusted until the ammeter current was reduced to zero. To account for fluctuations and establish measurement uncertainty, current readings were recorded for several voltage values on both sides of the identified stopping voltage.

The relationship between light intensity and electron energy was investigated by recording the stopping voltage for the three different apertures, all with the same 436 mm filter (Table 2).

Finally, the phototube was characterized by generating I-V curves for both the 2 mm and 8 mm apertures. This was achieved by recording the current as the applied voltage was swept from $-4.5\,\mathrm{V}$ to $3.0\,\mathrm{V}$ in $0.5\,\mathrm{V}$ increments, and then up to $29\,\mathrm{V}$ in $2\,\mathrm{V}$ increments (Figure 3).

3 Results

3.1 $V_{\mathbf{stop}}$ vs ν

In order to obtain an uncertainty for V_{stop}, a linear fit of the voltage vs. current data obtained for each wavelength (such as can be seen in Table 1) was made when the estimated uncertainty of the current was either subtracted or added to the current. The difference of the quotients of the fitting parameters yields the final uncertainty in V_{stop}, that is,

$$\delta V_{\text{stop}} = \frac{A_+}{B_+} - \frac{A_-}{B_-}.$$

Table 1: Voltages and currents around V_{stop} for the 365 nm filter.

Voltage (V)	Current (10^{-13} A)
-1.901	-25 ± 2
-1.896	-17 ± 5
-1.891	-11 ± 5
-1.886	-4 ± 6
-1.881	0 ± 3
-1.876	14 ± 4
-1.869	28 ± 5
-1.864	38 ± 3
-1.859	49 ± 4

Table 2: Light intensity vs. V_{stop}

aperture diameter (mm)	V_{stop} (V)
8	-1.316
4	-1.310
2	-1.283

By then making another fit, this time for V_{stop} vs ν, h and ϕ may be recovered:

$$\begin{aligned} V_{\text{stop}} &= A + B\nu \\ &= \frac{h}{e}\nu - \frac{\phi}{e} \\ &\implies \begin{cases} h = -eB \\ \phi = eA \end{cases} \end{aligned}$$

which came out to $h = (6.18 \pm 0.02) \times 10^{-34}\,\text{J}\,\text{s}$ and $\phi = (1.333 \pm 0.009)\,\text{eV}$. A plot of this can be seen in Figure 2

4 Conclusions

The measured h and an accepted h of $6.626 \times 10^{-34}\,\text{J}\,\text{s}$ yields a percent error of 6.7%. Additionally, Table 2 confirms that the energy of light is not dependent on its intensity.

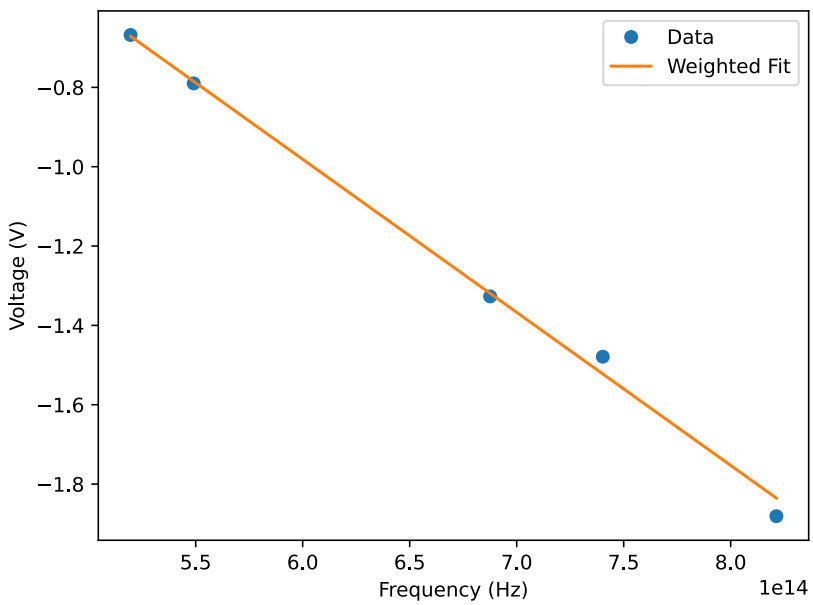

Data
Weighted Fit
−0.8
−1.0
−1.2
−1.4
−1.6
−1.8
Voltage (V)
5.5
6.0
6.5
7.0
7.5
8.0
Frequency (Hz)
1e14

Figure 2

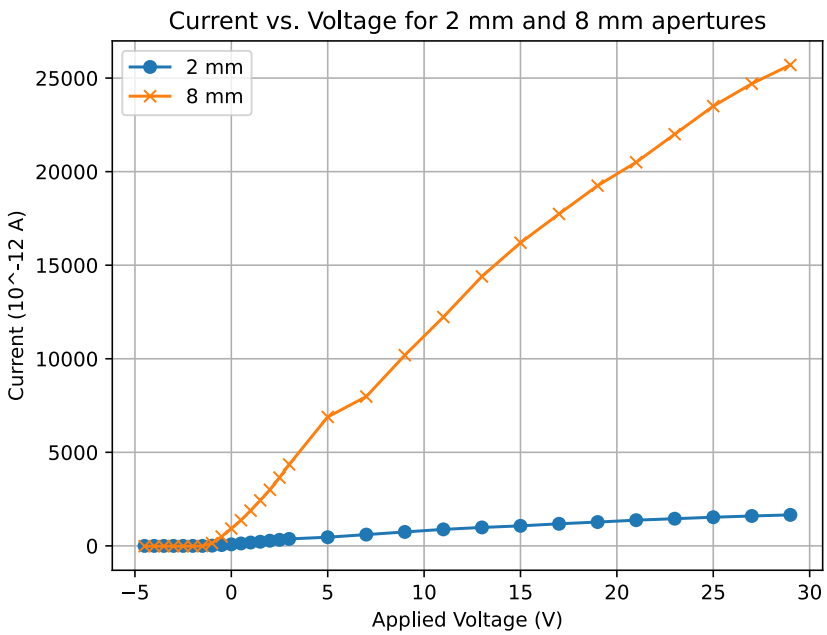

Current vs. Voltage for 2 mm and 8 mm apertures
2 mm
8 mm
Current (10^-12 A)
Applied Voltage (V)
25000
20000
15000
10000
5000
0
−5
0
5
10
15
20
25
30

Figure 3